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Education

University of Toronto

Sep 2024 - Present

Computer Science Internship Program & Statistics (GPA: 3.94 / 4.00)

Toronto, ON

- **Relevant Coursework:** Software Design, Data Structures, Machine Learning, Database Design, Systems Programming, Design of Interactive Computational Media, Methods of Data Analysis

Experience

University of Toronto

Dec 2025 – Present

Teaching Assistant

Mississauga, ON

- Conducted tutorials **1–2 times per week** for **40 students**, explaining quiz questions and **R code** approaches.
- Conducted live **data analysis coding demos**, explaining implementation logic alongside related course concepts.
- Assisted with course project development, including data collection and debugging R analysis code.

Beijing Century ChaoXing Information Technology Development Co., Ltd. •

Jul 2025 – Dec 2025

AI Agent Developer

Remote

- Built knowledge bases for **multiple smart-course agent projects** by consolidating/scraping instructor materials, university policies, and official website information into **6 domain-specific datasets** using **Python (pandas, NumPy)**.
- Standardized inconsistent information from instructors, academic offices, and campus systems, reducing repeated manual preprocessing and downstream coordination effort.
- Configured agent workflows and optimized prompts for **usability and citation grounding**, including support for an AI-powered **consultation and evaluation system** for medical students.

East China Normal University

Jun 2022 – Sep 2022

Student Research Assistant

Shanghai, China

- Led a student team to build a **Raspberry Pi–based Security Guard** using **face recognition for door access** and **voice interaction** for user registration, with **Python, OpenCV, and NumPy**; tested the prototype with **100+ community users** and won **Second Prize** at the **2nd Yangtze River Delta Artificial Intelligence Olympic Challenge (YAIOC)**.

Projects

Survey-Based Artwork Classification Model | Python, numpy, pandas, NLTK, scikit-learn

- Developed a **multi-class machine learning classification pipeline** for a structured survey dataset to classify artworks, covering data analysis, preprocessing, and model optimization.
- Cleaned open-text and numeric data using **pandas** and **NLTK**, converted text into **Naive Bayes features**, and combined them in a tuned **Random Forest** model, improving accuracy by **4%** and reach **92% validation accuracy**.

TMAP | HCI Research, Surveys, Experience Mapping, Interviewing, Empirical research, JavaScript, TypeScript

- Led a **formative HCI study** on campus-to-classroom wayfinding; synthesized **survey (N=38), observations, interviews, and background evidence** to pinpoint indoor-navigation breakdowns at UTSG.
- Translated **7 findings from field research** into measurable design requirements for prototyping.

WFH Burnout Data Analysis | R, Regression Modeling, Data Visualization, ggplot2

- Collaborated on an R-based analysis of a work-from-home burnout dataset, creating fitted trend plots, residual diagnostics, correlation heatmaps, and influence-analysis charts; contributed to regression modeling and model selection, helping build a final quadratic model with **Adjusted $R^2 = 0.9365$** .

LexiGo | Java, Spring Boot, REST APIs, JUnit5, Mockito, Maven, Swing, SQL

- Designed a language-learning app for vocabulary study and practice, reach **15 stars** on GitHub.
- Implemented reusable word-practice modules, contributing **15,000+ lines** of production-quality Java code.
- Led project setup and ongoing maintenance, including build/CI configuration and **code reviews**, resulting in **372 unit tests** and **82 merged pull requests** to ensure robust code quality.

Core Skills

Languages: Python, Java, C, C++, R, HTML5, CSS3, TypeScript, Assembly, SQL

Tools: Anaconda, LaTeX, Markdown, Linux, REST APIs, Git, GitLab, Docker